

- Can increase TCA levels; use with caution with TCAs or when switching from a TCA to sertraline
- Can cause a fatal “serotonin syndrome” when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped
- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing sertraline
- May displace highly protein-bound drugs (e.g., warfarin)
- Can rarely cause weakness, hyperreflexia, and incoordination when combined with sumatriptan or possibly with other triptans, requiring careful monitoring of patient
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)
- NSAIDs may impair effectiveness of SSRIs
- Via CYP450 2D6 inhibition, sertraline could theoretically interfere with the analgesic actions of codeine, and increase the plasma levels of some beta blockers and of atomoxetine
- Via CYP450 2D6 inhibition sertraline could theoretically increase concentrations of thioridazine and cause dangerous cardiac arrhythmias
- Via CYP450 3A4 inhibition, sertraline may increase the levels of alprazolam, buspirone, and triazolam

- Via CYP450 3A4 inhibition, sertraline could theoretically increase concentrations of certain cholesterol lowering HMG CoA reductase inhibitors, especially simvastatin, atorvastatin, and lovastatin, but not pravastatin or fluvastatin, which would increase the risk of rhabdomyolysis; thus, coadministration of sertraline with certain HMG CoA reductase inhibitors should proceed with caution
- Via CYP450 3A4 inhibition, sertraline could theoretically increase the concentrations of pimozide, and cause QTc prolongation and dangerous cardiac arrhythmias
- Via CYP450 3A4 inhibition, sertraline could theoretically increase the concentrations of pimozide, and cause QTc prolongation and dangerous cardiac arrhythmias
- False-positive urine immunoassay screening tests for benzodiazepine have been reported in patients taking sertraline due to a lack of specificity of the screening tests. False-positive results may be expected for several days following discontinuation of sertraline

Other Warnings/Precautions

- Add or initiate other antidepressants with caution for up to 2 weeks after discontinuing sertraline
- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent

- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is taking an MAOI
- If patient is taking pimozide
- If patient is taking thioridazine
- Use of sertraline oral concentrate is contraindicated with disulfiram due to the alcohol content of the concentrate
- If there is a proven allergy to sertraline

Special Populations

Renal Impairment

- No dose adjustment
- Not removed by hemodialysis

Hepatic Impairment

- Lower dose or give less frequently, perhaps by half

Cardiac Impairment

- Proven cardiovascular safety in depressed patients with recent myocardial infarction or angina
- Treating depression with SSRIs in patients with acute angina or following myocardial infarction may reduce cardiac events and improve survival as well as mood
- Should be used with caution in patients with risk factors for QTc prolongation

Elderly

- Some patients may tolerate lower doses and/or slower titration better
- Risk of SIADH with SSRIs is higher in the elderly
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Approved for use in OCD
- Ages 6–12: initial dose 25 mg/day
- Ages 13 and up: adult dosing
- Long-term effects, particularly on growth, have not been studied

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Not generally recommended for use during pregnancy, especially during first trimester

- Nonetheless, continuous treatment during pregnancy may be necessary and has not been proven to be harmful to the fetus
- At delivery there may be more bleeding in the mother and transient irritability or sedation in the newborn
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child
- For many patients this may mean continuing treatment during pregnancy
- Exposure to SSRIs early in pregnancy may be associated with increased risk of septal heart defects (absolute risk is small)
- SSRI use beyond the 20th week of pregnancy may be associated with increased risk of pulmonary hypertension in newborns, although this is not proven
- Exposure to SSRIs late in pregnancy may be associated with increased risk of gestational hypertension and preeclampsia
- Neonates exposed to SSRIs or SNRIs late in the third trimester have developed complications requiring prolonged hospitalization, respiratory support, and tube feeding; reported symptoms are consistent with either a direct toxic effect of SSRIs and SNRIs or, possibly, a drug discontinuation syndrome, and include respiratory distress, cyanosis, apnea, seizures, temperature instability, feeding

difficulty, vomiting, hypoglycemia, hypotonia, hypertonia, hyperreflexia, tremor, jitteriness, irritability, and constant crying

- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- Trace amounts may be present in nursing children whose mothers are on sertraline
- Sertraline has shown efficacy in treating postpartum depression
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Patients with atypical depression (hypersomnia, increased appetite)
- Patients with fatigue and low energy
- Patients who wish to avoid hyperprolactinemia (e.g., pubescent children, girls and women with galactorrhea, girls and women with unexplained amenorrhea, postmenopausal women who are not taking estrogen replacement therapy)
- Patients who are sensitive to the prolactin-elevating properties of other SSRIs (sertraline is the one SSRI that generally does not elevate prolactin)

Potential Disadvantages

- Initiating treatment in anxious patients with some insomnia
- Patients with comorbid irritable bowel syndrome
- Can require dosage titration

Primary Target Symptoms

- Depressed mood
- Anxiety
- Obsessions, compulsions

- Sleep disturbance, both insomnia and hypersomnia (eventually, but may actually cause insomnia, especially short-term)
- Panic attacks, avoidant behavior, re-experiencing, hyperarousal

Pearls

- * May be a type of “dual action” agent with both potent serotonin reuptake inhibition and less potent dopamine reuptake inhibition, but the clinical significance of this is unknown
- Cognitive and affective “flattening” may theoretically be diminished in some patients by sertraline’s dopamine reuptake blocking properties
- * May be a first-line choice for atypical depression (e.g., hypersomnia, hyperphagia, low energy, mood reactivity)
- Best documented cardiovascular safety of any antidepressant, proven safe for depressed patients with recent myocardial infarction or angina
- May bind to sigma 1 receptors, enhancing sertraline’s anxiolytic actions
- Can have more gastrointestinal effects, particularly diarrhea, than some other antidepressants
- May be more effective treatment for women with PTSD or depression than for men with PTSD or depression, but the clinical significance of this is unknown

- SSRIs may be less effective in women over 50, especially if they are not taking estrogen
- SSRIs may be useful for hot flushes in perimenopausal women
- For sexual dysfunction, can augment with bupropion, sildenafil, vardenafil, tadalafil, or switch to a non-SSRI such as bupropion or mirtazapine
- Some postmenopausal women's depression will respond better to sertraline plus estrogen augmentation than to sertraline alone
- Nonresponse to sertraline in elderly may require consideration of mild cognitive impairment or Alzheimer disease
- Not as well tolerated as some SSRIs for panic, especially when dosing is initiated, unless given with co-therapies such as benzodiazepines or trazodone
- Relative lack of effect on prolactin may make it a preferred agent for some children, adolescents, and women
- Some evidence suggests that sertraline treatment during only the luteal phase may be more effective than continuous treatment for patients with PMDD

Suggested Reading

Cipriani A , La Ferla T , Furukawa TA , et al. Sertraline versus other antidepressive agents for depression . *Cochrane Database Syst Rev* 2010

;14 (4):CD006117 .

DeVane CL , Liston HL , Markowitz JS . Clinical pharmacokinetics of sertraline . *Clin Pharmacokinet* 2002 ;41 :1247–66 .

Flament MF , Lane RM , Zhu R , Ying Z . Predictors of an acute antidepressant response to fluoxetine and sertraline . *Int Clin Psychopharmacol* 1999 ;14 :259–75 .

Khouzam HR , Emes R , Gill T , Raroque R . The antidepressant sertraline: a review of its uses in a range of psychiatric and medical conditions . *Compr Ther* 2003 ;29 :47 –53 .

McRae AL , Brady KT . Review of sertraline and its clinical applications in psychiatric disorders . *Expert Opin Pharmacother* 2001 ;2 :883–92 .

Sildenafil

Therapeutics

Brands

- ViagraRevatio
- *see index for additional brand names*

Generic?

Yes

Class

- Phosphodiesterase-5 (PDE-5) inhibitor

Commonly Prescribed for

(bold for FDA approved)

- **Erectile dysfunction (Viagra)**
- **Pulmonary arterial hypertension (Revatio)**

How the Drug Works

- PDE-5 is responsible for degradation of cyclic guanosine monophosphate (cGMP); cGMP produces smooth muscle relaxation in the corpus cavernosum and allows inflow of blood

- Inhibition of PDE-5 causes increased levels of cGMP in the corpus cavernosum, resulting in smooth muscle relaxation and inflow of blood to the corpus cavernosum
- Sildenafil has no direct relaxant effects on isolated human corpus cavernosum, and at recommended doses has no effect in the absence of sexual stimulation

How Long Until It Works

- Maximum observed plasma concentrations are reached within 30–120 minutes (median 60 minutes) of oral dosing

If It Works

- Increases ability to achieve and sustain an erection sufficient for satisfactory sexual activity

If It Doesn't Work

Can increase dose to 100 mg if not already tried

If there is no improvement, discontinue use



Best Augmenting Combos for Partial Response or Treatment Resistance

- None known

Tests

None for healthy individuals

Side effects

How Drug Causes Side Effects

- Inhibition of PDE-5 in tissues not associated with erectile dysfunction

Notable Side Effects

Headache

Flushing

Dyspepsia (indigestion, diarrhea)



Life-Threatening or Dangerous Side Effects

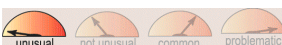
- Sudden severe loss of vision and/or blurred vision
- Sudden decrease or loss of hearing, ringing in ears
- Erection that is painful or lasts longer than 4 hours

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected

What to Do About Side Effects

- Reduce dose to 25 mg

Best Augmenting Agents for Side Effects

Over-the-counter medications for headache or dyspepsia

Sildenafil itself is used to treat erectile dysfunction as a side effect of medications, notably paroxetine

Dosing and use

Usual Dosage Range

- Erectile dysfunction: 25–100 mg as needed, approximately 1 hour before sexual activity (Viagra)
- Pulmonary arterial hypertension (tablet and oral suspension): 5 mg or 20 mg three times a day, 4–6 hours apart

Dosage Forms

- Tablet 25 mg, 50 mg, 100 mg (Viagra)
- Tablet 5 mg, 20 mg (Revatio)
- Injectable solution 10 mg/12.5 mL (Revatio)
- Oral suspension 10 mg/mL (Revatio)

How to Dose

- 50 mg as needed, recommended approximately 1 hour before sexual activity; however, may be taken anywhere from 30 minutes to 4 hours before sexual activity



Dosing Tips

Maximum recommended dosing frequency is once per day

Can adjust dose to a maximum of 100 mg or decrease to 25 mg depending on efficacy and tolerability

Can be taken with or without food

Overdose

- Similar symptoms as at usual doses, but incidence rates and severity are increased

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- When used for erectile dysfunction, sildenafil is administered as needed

Pharmacokinetics

- Maximum observed plasma concentrations are reached within 30–120 minutes (median 60 minutes) of oral dosing
- Metabolized to an active metabolite, primarily by CYP450 3A4 (major route) and CYP450 2C9 (minor route)
- Parent drug and its active metabolite have approximately 4-hour terminal half-lives
- Absorption is delayed if taken with a high-fat meal



Drug Interactions

CYP450 3A4 inhibitors (e.g., ritonavir, ketoconazole, itraconazole, erythromycin, fluvoxamine, fluoxetine) can increase plasma levels of sildenafil

In patients taking ritonavir, maximum single dose is 25 mg in a 48-hour period

In patients taking erythromycin or strong CYP450 3A4 inhibitors, consider a starting dose of 25 mg

Additive blood pressure lowering effects may occur with concomitant use of nitrates, alpha blockers, and antihypertensives

If coadministered with an alpha blocker, the patient should be stable on the alpha blocker prior to initiating sildenafil; starting dose of sildenafil should be 25 mg



Other Warnings/Precautions

Patients should not use sildenafil if sexual activity is inadvisable due to cardiovascular status

Patients should seek emergency treatment if an erection lasts more than 4 hours

Use with caution in patients who are predisposed to priapism or who have anatomical deformation of the penis

Patients should stop taking sildenafil and seek medical attention if sudden loss of vision occurs in one or both eyes; sildenafil should be used only with caution in patients with a history of non-arteritic anterior ischemic optic neuropathy (NAION)

Use with caution in patients with retinitis pigmentosa

Patients should stop taking sildenafil and seek medical attention if there is a sudden decrease or loss of hearing

Use cautiously with alpha blockers or antihypertensives, as it may lead to hypotension

Decreased blood pressure, syncope, and prolonged erection may occur at higher sildenafil exposures

Not recommended for use in combination with other PDE-5 inhibitors

Do Not Use

- If patient is using organic nitrates in any form
- If patient is taking a guanylate cyclase inhibitor
- If there is a proven allergy to sildenafil

Special populations

Renal Impairment

- No adjustment for mild to moderate impairment
- For patients with severe impairment, consider a starting dose of 25 mg

Hepatic Impairment

- Consider a starting dose of 25 mg

Cardiac Impairment

- Elicits vasodilatory properties, resulting in mild and transient decreases in blood pressure
- Treatment for erectile dysfunction generally should not be instituted in men for whom sexual activity is inadvisable because of their underlying cardiovascular status
- There are no controlled clinical data on the safety or efficacy of sildenafil in patients who have suffered a myocardial infarction, stroke, or life-threatening arrhythmia within the last 6 months; have resting hypotension (BP <90/50 mmHg) or hypertension (BP >170/110 mmHg); or have cardiac failure or coronary artery disease causing unstable angina; use only with caution in these patients
- Use with caution in patients with left ventricular outflow obstruction (e.g., aortic stenosis, idiopathic hypertrophic subaortic stenosis) and those with severely impaired autonomic control of blood pressure

Elderly

- Some patients may tolerate lower doses better
- Consider a starting dose of 25 mg



Children and Adolescents

- Not recommended for use in pediatric patients
- Safety and efficacy have not been established



Pregnancy

- Viagra is not indicated for use in women
- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Animal studies did not show teratogenicity or adverse developmental outcomes when administered during organogenesis at doses up to 16 and 32 times the maximum recommended human dose

Breast Feeding

- Some drug is found in mother's breast milk

The art of psychopharmacology

Potential Advantages

- Can be taken as needed

Potential Disadvantages

- Patients with cardiovascular disease
- Patients who take concomitant medications that may interact with sildenafil's pharmacokinetic and pharmacodynamic properties

Primary Target Symptoms

- Erectile dysfunction



Pearls

Patients should be counseled that sildenafil offers no protection against sexually transmitted diseases



Suggested Reading

Andersson KE. Andersson KE. PDE5 inhibitors –pharmacology and clinical applications 20 years after sildenafil discovery . *Br J Pharmacol* 2018 ;**175** (13):2554 –65 .

Carson CC Carson CC . Long-term use of sildenafil . *Expert Opin Pharmacother* 2003 ;**4** (3):397 –405 .

Sodium oxybate

Therapeutics

Brands

- Xyrem
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: GABA receptor agonist
- CNS depressant; GABA-B receptor partial agonist

Commonly Prescribed for

(bold for FDA approved)

- **Reducing excessive sleepiness in patients with narcolepsy (ages 7 and older)**
- **Cataplexy in patients with narcolepsy (ages 7 and older)**
- Fibromyalgia
- Chronic pain/neuropathic pain

How the Drug Works

- Gamma hydroxybutyrate (GHB) is an endogenous putative neurotransmitter synthesized from its parent compound, GABA; sodium oxybate is the sodium salt of GHB and is administered exogenously
- Has agonist actions at GHB receptors and partial agonist actions at GABA-B receptors
- Improves slow-wave sleep at night, presumably leaving patients better rested and more alert during the day

How Long Until It Works

- Can immediately reduce daytime sleepiness
- Can take several days to optimize dosing and clinical improvement

If It Works

- Improves daytime sleepiness and may improve fatigue in patients with narcolepsy
- Reduces frequency of cataplexy attacks
- May improve sleep physiology and subjective experience of pain and fatigue in patients with fibromyalgia and other chronic pain conditions

If It Doesn't Work

- Increase dose

- Consider an alternative treatment

Best Augmenting Combos for Partial Response or Treatment Resistance

- * Sodium oxybate is itself often used as an augmenting therapy to stimulants, modafinil, or armodafinil for excessive sleepiness in narcolepsy
- * The majority of patients taking sodium oxybate in clinical trials for narcolepsy and cataplexy were taking a concomitant stimulant
- Often used as an augmenting agent for fibromyalgia with SNRIs (e.g., duloxetine, milnacipran) and alpha 2 delta ligands (e.g., gabapentin, pregabalin), but has not been well studied in combination with these agents

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Unknown
- CNS side effects presumably due to excessive CNS actions on various neurotransmitter systems

Notable Side Effects

- Headache, dizziness, sedation
- Nausea, vomiting
- Enuresis



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken in overdose
- Neuropsychiatric events (psychosis, depression, paranoia, agitation)
- Confusion and wandering at night (unclear if this is true somnambulism)

Weight Gain



- Reported but not expected
- May cause weight loss

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Lower the dose